

4. Career Development (28)

AP CS classes Niki Lauda Assignment:

Nikki Lauda became one of the most successful Formula 1 drivers in history not simply because he could drive a race car, but because he understood the factors that affected performance. Rather than focusing only on driving, he asked questions about weight, materials, reliability, and design.

In computer science, programming languages such as Java and Python, as well as AI tools, are just that: **tools**. The most successful computer scientists and software engineers first understand the problem, identify important variables and constraints, and then use technology to build solutions.

This activity asks you to think critically about the relationship between problem-solving and programming.

Video: **RUSH (2013) | Niki Lauda joins to BRM team (5min)**

[Niki_Lauda_Exit_Ticket_APCSA.docx](#)Download [Niki_Lauda_Exit_Ticket_APCSA.docx](#)

Instructions

1. Read each question carefully.
2. Answer all questions completely and thoughtfully.
3. Focus on explaining your thinking, not on writing code.
4. Use examples from real-world situations when appropriate.
5. Submit your completed document

AI Expectations

You may use AI to clarify vocabulary or concepts if needed.

However, your responses must reflect **your own thinking and reasoning**.

Remember: AI can generate code, but it cannot automatically determine the most important problem to solve, or the needs of the user.

Your goal is to demonstrate that you understand the problem-solving process, not simply that you can generate an answer.

The screenshot shows a Canvas LMS interface. The top navigation bar includes the user's name 'PRCS NIKKI Lauda', the course 'P3-AP Computer Sci A-Torres', and a progress bar showing '10/10' and '88.30 / 100 (87%)'. The assignment title is 'Problem Solving Before Programming'. The due date is 'Sep 4 at 11:59 PM'. The assignment is graded, with an average of '88.30 / 100 (87%)'. The student's name is 'Angel Valladares-Ortez'. The course is 'AP CSA / AP CSP'. The assignment is dated '8 September 2026'. The assignment description reads: 'Read: Niki Lauda became successful not only because he could drive a race car, but because he understood the factors affecting performance. He asked questions about weight, materials, reliability, and design. In computer science, Java, Python, and AI are tools. Great problem-solvers first understand the problem, identify important variables, and then use tools to build solutions. RUSH (2013) | Niki Lauda joins to BRM team - Full scene | Kinoman'. The assignment is divided into two sections: '1. Find the Problem' and '2. Important Variables'. Section 1 includes a list of tasks: 'Create classes and methods', 'Build the user interface', 'Understand why students cannot find parking' (checked), and 'Ask AI to create the app'. Section 2 includes a prompt: 'Suppose you are designing a flight tracking system. List four important variables'. The right sidebar shows the 'Submitted Files' section with a file 'Niki_Lauda_Exit_Ticket_APCSA.do' and a '58%' completion status. The 'Assessment' section shows a 'Grade (90 / 100)' of '90%' and a 'Status' of 'Excellent'. The 'Assignment Comments' section shows a comment from 'Aldrin Torres' dated 'Sep 4 at 11:59 PM' with the text: 'thanks for your comments, Programming is a tool, Understanding the problem is the profession.' The bottom of the sidebar shows a 'Paragraph' dropdown menu.

PRCS NIKKI Lauda
Due: Sep 4 at 11:59 PM - P3-AP Computer Sci A-Torres
Graded Average Students
10/10 88.30 / 100 (87%) 14/18
Submitted Files: (click to load)
Niki_Lauda_Exit_Ticket_APCSA.do
58% Student Viewed Document: Sep 9 at 4:2 PM
Assessment
Grade (90 / 100) Status
90%
Days
Due: Sep 4 at 1
Assignment Comments
Excellent
thanks for your comments,
Programming is a tool,
Understanding the problem is
the profession.
Aldrin Torres,
Paragraph

Problem Solving Before Programming

Name: Angel Valladares-Ortez Date: 8 September 2026
Course: AP CSA / AP CSP

Read: Niki Lauda became successful not only because he could drive a race car, but because he understood the factors affecting performance. He asked questions about weight, materials, reliability, and design. In computer science, Java, Python, and AI are tools. Great problem-solvers first understand the problem, identify important variables, and then use tools to build solutions.
[RUSH \(2013\) | Niki Lauda joins to BRM team - Full scene | Kinoman](#)

1. Find the Problem

A school wants an app to help students find parking spaces. Before writing any Java code, what should the team do first?

- ☐ Create classes and methods
- ☐ Build the user interface
- ☒ Understand why students cannot find parking
- ☐ Ask AI to create the app

Why? It would be futile to try and start building code or prompting AI if you do not even know what you are doing it for. You need to first understand the details of the problem you are trying to address, such as why it occurs, before you can start thinking of how to solve it.

2. Important Variables

Suppose you are designing a flight tracking system. List four important variables

Gaming Industry Research:

Maximum: 8 Slides Individual Assignment

Driving Question

If you were going to build a game, what type of game would you create and why would it have the best chance of success?

Assignment Goal

Research the game industry, explore different game categories, and identify a game type that interests you. Use industry information to justify why your selected game genre has market potential, career opportunities, and manageable ethical or legal concerns.

Games that interest me



Game	Genre	Why I like it
Genshin Impact	Exploration/ Adventure	Open world with beautiful graphics
Pokémon Go	Adventure	Exploration, Collection
Among US	Social deduction, hidden roles	Mystery, multiplayer, fun
Candy Crush	Puzzle	Quick puzzles, low stress
Subway Surfers	Action	Interactive

I find Genshin the most interesting because of its flexible gameplay, story and overall exploration atmosphere

Submitted: Aug 21 at 7:06 PM

Word Count: 450

Submitted Files: (click to load)

[Game Design Industry Research Project.pptx](#)

Assessment

Grade (95 / 100)

Status

95%

None

Assignment Comments



good AMA citing, thanks
it will be more appealing to your
market opportunity if you provide
evidence of your points: example:
[Action Games Market Size, Share & Industry Forecast](#)



, Aug 22 at 9:17 AM